

Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/23

Paper 2 Further Pure Mathematics 2

May/June 2022

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

- 1 Find the roots of the equation $z^3 = 7\sqrt{3} - 7i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi \leq \theta < \pi$. [5]

[illegible]

- 2 (a) Find the coefficient of x^2 in the Maclaurin's series for $-\ln \cos x$. [4]

[illegible]

- (b)** Find the length of the arc of the curve with equation $y = -\ln \cos x$ from the point where $x = 0$ to the point where $x = \frac{1}{4}\pi$. [4]

[illegible]

- (b) Use the characteristic equation of $\mathbf{A}$ to show that $\mathbf{A}^{-1} = p\mathbf{A}^2 + q\mathbf{I}$, where p and q are constants to be determined. [3]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

4 It is given that

$$x = -t + \tan^{-1} t \quad \text{and} \quad y = t + \sinh^{-1} t.$$

(a) Show that $\frac{dy}{dx} = -\frac{t^2 + 1 + \sqrt{t^2 + 1}}{t^2}$. [4]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal dotted lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

(b) Find the value of $\frac{d^2y}{dx^2}$ when $t = \frac{3}{4}$.

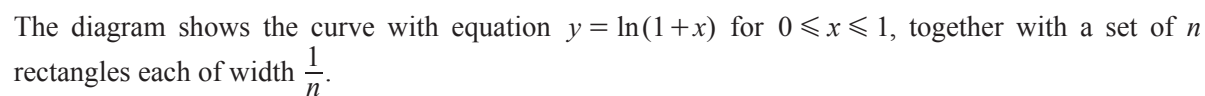
[5]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.

- 5 Find the solution of the differential equation

$$x(x+7)\frac{dy}{dx} + 7y = x$$

for which $y = 7$ when $x = 1$. Give your answer in the form $y = f(x)$. [9]


$$U_n = \frac{1}{n} \ln \frac{(2n)!}{n!} - \ln n. \quad [4]$$
[illegible]

(b) Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 \ln(1+x) dx$. [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

(c) By simplifying $U_n - L_n$, show that $\lim_{n \rightarrow \infty} (U_n - L_n) = 0$. [2]

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and extend across the entire width of the page. There is no text or other markings on the paper.

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(b) Deduce the exact value of x for which $y = 0$. Give your answer in logarithmic form. [3]

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- 8 (a)** Find $\int \sin \theta \cos^n \theta \, d\theta$, where $n \neq -1$. [2]

Let $I_{m,n} = \int_0^{\frac{1}{2}\pi} \sin^m \theta \cos^n \theta \, d\theta$.

- (b)** Show that, for $m \geq 2$ and $n \geq 0$,

$$I_{m,n} = \frac{m-1}{m+n} I_{m-2,n}. \quad [5]$$

- (c) By considering the binomial expansion of $\left(z + \frac{1}{z}\right)^5$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that

$$\cos^5 \theta = a \cos 5\theta + b \cos 3\theta + c \cos \theta,$$

where a , b and c are constants to be determined. [5]

- (d) Using the results given in parts (b) and (c), find the exact value of $I_{2,5}$. [4]

[illegible]

This image shows a full page of a document template designed for handwriting practice. It features a series of evenly spaced, horizontal dashed lines across the entire width of the page. The background is plain white, providing a clear guide for letter height and placement. There are no margins, text, or other markings present.

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